MATH 110 — Week 9

Notes Chapters 16–17; Homework 9

§1 Trace of matrices and operators

The question. Which simple quantity remains unchanged under a change of basis? The matrices A and $S^{-1}AS$ can have different entries, but their diagonal sums agree. For example $\operatorname{diag}(1,2)$ has sum 3 in every basis.

Tools from earlier weeks.

- Weeks 3–4: $(AB)_{ij} = \sum_k a_{ik}b_{kj}$ follows by substituting basis-vector images; a basis change gives $S^{-1}AS$, with S invertible.
- Week 8: complex Jordan form has one block $J_r(\lambda)$ per chain. Each contributes dimension r to its generalized eigenspace and one eigenvector direction; the largest block length is the exponent of $x-\lambda$ in m_T.
- Week 8: $\chi_T = \prod_\lambda (x-\lambda)^{d_\lambda}$, where d_λ is algebraic multiplicity. The multiplicities sum to dimension by the generalized eigenspace direct sum.

Definition (Notes Definition 41; LADR 8.47)**.** The trace of an n-by-n matrix A is $\operatorname{tr} A = \sum_{j=1}^n a_{jj}$. An empty matrix has trace 0.

Example (Notes Chapter 16; LADR 8.48)

The matrix $\left(\begin{smallmatrix}1&2&3\\4&5&6\\7&8&9\end{smallmatrix}\right)$ has trace $1+5+9=15$. The textbook matrix $\left(\begin{smallmatrix}3&-1&-2\\3&2&-3\\1&2&0\end{smallmatrix}\right)$ has trace $3+2+0=5$.

Proposition (Notes Proposition 78; LADR 8.49)

For A of size m-by-n and B of size n-by-m, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Where the proof comes from

Want. Compare diagonal sums even when the two products have different sizes.
Know. The diagonal entries of each product are sums over the other index.
Key step. Reverse the order of a finite double sum and commute scalar factors.

Proof. 1. Expand $\operatorname{tr}(AB) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ji}$.
2. Reverse the finite sums and use field commutativity: this equals $\sum_{j=1}^n \sum_{i=1}^m b_{ji}a_{ij} = \operatorname{tr}(BA)$.

□

Proposition (Notes Definition 42, well-definedness; LADR 8.50)

Similar matrices have the same trace. Thus the trace of a representing matrix depends only on the operator.

Where the proof comes from

Want. Define trace without selecting a preferred basis.
Know. A basis change gives $S^{-1}AS$, and trace interchanges the factors of a product.
Key step. Move the leftmost S^{-1} to the right, where it cancels S.

Proof. 1. For invertible S, Proposition 78 gives $\operatorname{tr}(S^{-1}AS) = \operatorname{tr}(ASS^{-1}) = \operatorname{tr} A$.
2. Every pair of representing matrices is related by such a basis change (Week 4), proving independence of the basis.

□

Definition (Notes Definition 42; LADR 8.51)**.** For $T \in L(V)$ on finite-dimensional V, define $\operatorname{tr} T = \operatorname{tr} M_{\mathcal{B}}(T)$ in any basis $\mathcal{B}$; the preceding result makes this well defined.

Proposition (LADR 8.56: linearity and cyclic products)
Trace is a linear functional on $L(V)$: $\operatorname{tr}(aS + bT) = a \operatorname{tr} S + b \operatorname{tr} T$. Also $\operatorname{tr}(ST) = \operatorname{tr}(TS)$, and $\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$ for operators on the same space.

Where the proof comes from

Want. Simplify traces using algebra before finding eigenvalues.
Know. Addition and scaling of matrices act entrywise; Proposition 78 interchanges two factors.
Key step. Add diagonal entries and group a longer product as two factors.

Proof. 1. In any common basis, $\sum_j (aS + bT)_{jj} = a \sum_j S_{jj} + b \sum_j T_{jj}$, proving linearity.
2. Proposition 78 applies to representing matrices and proves $\operatorname{tr}(ST) = \operatorname{tr}(TS)$.
3. Apply it to $(AB)C$ to get $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$, and to $A(BC)$ to get $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$. Associativity justifies the regrouping.

□

Proposition (Notes Proposition 79; LADR 8.52)
For a complex finite-dimensional operator, $\operatorname{tr} T = \sum_\lambda d_\lambda \lambda$, the sum of its eigenvalues with algebraic multiplicity.

Where the proof comes from

Want. Recover an eigenvalue sum without solving individual eigenvalues.
Know. Trace permits any basis and Jordan form puts eigenvalues on the diagonal.
Key step. Compute in a Jordan basis and count block dimensions.

Proof. 1. Choose the Jordan basis provided by Week 8. Each block $J_r(\lambda)$ contributes $r\lambda$ to the diagonal sum.
2. Sum over blocks; the total block size for λ is its algebraic multiplicity d_λ (Week 8).

□

Example (Notes Example 64; LADR 8.53) 1. For $A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{smallmatrix}\right)$, $\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} A = 1 + 2 + 9 = 12$, with roots repeated according to multiplicity. No separate root formulas are needed.
2. For the textbook matrix $B = \left(\begin{smallmatrix} 3 & -1 & -2 \\ 3 & 2 & -3 \\ 1 & 2 & 0 \end{smallmatrix}\right)$, the eigenvalues are $1, 2 + 3i, 2 - 3i$, each once: direct multiplication verifies corresponding eigenvectors $(1, 0, 1)$, $(i, 1 + i, 1)$, $(-i, 1 - i, 1)$. Distinct eigenvalues give an eigenbasis (Week 7), so each multiplicity is 1 and the characteristic polynomial is $(x - 1)((x - 2)^2 + 9)$.
3. Their sum is $1 + (2 + 3i) + (2 - 3i) = 5 = \operatorname{tr} B$, agreeing with the diagonal calculation.

Proposition (Notes Chapter 16; LADR 8.54: characteristic coefficient)
For $n = \dim V \geq 1$ over $\mathbb{C}$, the coefficient of x^{n-1} in χ_T is $-\operatorname{tr} T$.

Where the proof comes from

Want. Read trace directly from a characteristic polynomial.
Know. χ_T is the product of n linear factors, with multiplicity.
Key step. A term of degree $n-1$ selects a constant from exactly one factor.

Proof. 1. Write $\chi_T = \prod_{j=1}^{n}(x - \lambda_j)$ with repetitions. To obtain x^{n-1}, select $-\lambda_j$ from one factor and x from the rest.
2. Summing these contributions gives $-\sum_j \lambda_j = -\operatorname{tr} T$ by Proposition 79.

□

Proposition (Notes Proposition 80; LADR 8.57)
The commutator $ST - TS$ has trace zero. On a nonzero finite-dimensional real or complex space it cannot equal I.

Where the proof comes from

Want. Rule out an operator equation without solving for its unknown entries.
Know. Trace is linear and cyclic, while $\operatorname{tr} I = (\dim V)1_F$.
Key step. Compare the trace of each side.

Proof. 1. Linearity and Proposition 78 give $\operatorname{tr}(ST - TS) = \operatorname{tr}(ST) - \operatorname{tr}(TS) = 0$.
2. Over $\mathbb{R}$ or $\mathbb{C}$ on nonzero V, $\operatorname{tr} I = \dim V > 0$. Thus $ST - TS \neq I$. The field restriction ensures this positive integer is nonzero as a scalar.

□

Compute a nonzero commutator

1. Take $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Then $AB = \operatorname{diag}(1, 0)$ and $BA = \operatorname{diag}(0, 1)$.
2. Hence $AB - BA = \operatorname{diag}(1, -1) \neq 0$, but its trace is $1 - 1 = 0$.

Shortcuts

- Choose a basis adapted to invariant parts: diagonal or Jordan blocks often make trace immediate.
- To find the sum of all complex eigenvalues, add diagonal entries; root solving is unnecessary.
- Cyclically rotate factors inside trace to cancel an adjacent inverse; this follows from Proposition 78.

Traps

- Trace zero does not imply nilpotent: $\operatorname{diag}(1, -1)$ has trace 0, but its square is I.
- Cyclic rotation is not arbitrary reordering: for $A = E_{12}, B = E_{21}, C = E_{11}$, $\operatorname{tr}(ABC) = 1$ while $\operatorname{tr}(ACB) = 0$. Here E_{ij} has a 1 at position (i, j) and zeros elsewhere.

§2 Inner products and examples

The question. What extra structure assigns a squared length to every vector? On $\mathbb{C}^2$, $\langle z, w\rangle = z_1\overline{w_1} + z_2\overline{w_2}$ gives $\langle (i,1),(i,1)\rangle = 2$, whereas omitting conjugates gives 0.

Definition (Notes Definition 43; LADR 6.1–6.4)**.** For $F = \mathbb{R}$ or $\mathbb{C}$, an inner product on V is a function $\langle u, v\rangle \in F$ with $\langle v, v\rangle > 0$ for $v \neq 0$, linearity in the first argument, and conjugate symmetry $\langle u, v\rangle = \overline{\langle v, u\rangle}$. An inner-product space is a vector space equipped with such a function. On $\mathbb{R}$, conjugation has no effect.

Proposition (Notes Definition 43, sesquilinearity; LADR 6.6)

An inner product satisfies $\langle 0, v\rangle = \langle v, 0\rangle = 0$ and conjugate linearity in the second argument: $\langle u, av + bw\rangle = \overline{a}\langle u, v\rangle + \overline{b}\langle u, w\rangle$. Also $\langle v, v\rangle$ is real, nonnegative, and zero exactly when $v = 0$.

Where the proof comes from

Want. Compute with both argument positions while keeping conjugates correct.
Know. The first position is linear and swapping positions conjugates.
Key step. Swap, use first-position linearity, and conjugate back.

Proof. 1. Linearity gives $\langle 0, v\rangle = 0\langle 0, v\rangle = 0$; conjugate symmetry gives $\langle v, 0\rangle = 0$.
2. $\langle u, av + bw\rangle = \overline{\langle av + bw, u\rangle} = \overline{a\langle v, u\rangle + b\langle w, u\rangle}$, which is the asserted formula.
3. Symmetry at $u = v$ gives a number equal to its conjugate, hence real. Positivity at nonzero vectors and the zero equation give the remaining claims.

□

Example (Notes Example 65; LADR 6.3(a)–(b))

For positive real $c_1, \ldots, c_n$, $\langle z, w\rangle_c = \sum_j c_j z_j \overline{w_j}$ is an inner product; the standard dot product has every $c_j = 1$.

1. Distribution of finite sums gives first-position linearity. Conjugation of the sum interchanges z, w because the c_j are real.
2. $\langle z, z\rangle_c = \sum_j c_j |z_j|^2 \geq 0$; if $z \neq 0$, one positive summand makes the sum positive. For $c = (1, 2)$ and $z = (1, i)$ its diagonal value is $\langle z, z\rangle_c = 1 + 2 = 3$.

Example (Notes Example 66; LADR 6.3(c))

For continuous F-valued functions on $[a, b]$ with $a < b$, $\langle f, g\rangle = \int_a^b f(x)\overline{g(x)}\,dx$ is an inner product, called the L^2 product. This includes the notes' interval $[0, 1]$ and textbook interval $[-1, 1]$.

1. Linearity of the integral and conjugation give first-position linearity and conjugate symmetry; these elementary integration facts are cited from calculus.
2. If continuous f is nonzero at some point, continuity makes $|f|^2$ bounded below by a positive number on a subinterval of positive length. Hence $\int_a^b |f|^2 > 0$.
3. For example on $[0, 1]$, $\langle 1, x\rangle = 1/2$ and $\langle x, x\rangle = 1/3$. On unbounded intervals one must restrict the function space so all these integrals exist.

Example (LADR 6.3(d): a derivative inner product)

On real polynomials define $\langle p, q\rangle = p(0)q(0) + \int_{-1}^1 p'(x)q'(x)\,dx$.

1. Evaluation, differentiation and integration are linear, so this is bilinear and symmetric.
2. If $\langle p, p\rangle = 0$, both nonnegative terms are zero. By the continuity argument above, $p' = 0$, so p is constant; $p(0) = 0$ then gives $p = 0$. Thus it is positive definite.
3. For example $\langle 1, 1\rangle = 1$, $\langle x, x\rangle = 2$ and $\langle x^2, x^2\rangle = \int_{-1}^1 4x^2\,dx = 8/3$.

Example (LADR 6.3(e): an unbounded interval)

On real polynomials, $\langle p, q\rangle = \int_0^\infty p(x)q(x)e^{-x}\,dx$ is an inner product.

1. Cited calculus tools are integration by parts and $x^k e^{-x} \to 0$ as $x \to \infty$ for fixed nonnegative integer k. They give $\int_0^\infty x^k e^{-x}\,dx = k!$: start at $k = 0$ and integrate by parts to reduce to $k - 1$.
2. More explicitly the truncated nonnegative integrals are bounded above by k times the preceding complete integral, so their increasing limit exists. The boundary term vanishes by the cited limit. Finite polynomial sums therefore converge absolutely after multiplication by e^{-x}.
3. Linearity and symmetry follow from integration. The weight is strictly positive, and any nonzero polynomial is nonzero on a subinterval, making its squared integral positive.
4. The moments give $\langle 1, x\rangle = 1$, $\langle x, x\rangle = 2$ and $\langle x^2, x^2\rangle = 24$.

Reject a proposed inner product

1. The real formula $x_1y_1 - x_2y_2$ gives $\langle e_2, e_2\rangle = -1$, violating positivity.
2. The formula $(x_1 + x_2)(y_1 + y_2)$ gives diagonal value 0 to $(1, -1) \neq 0$, violating positive definiteness.
3. The unconjugated complex formula $z_1w_1 + z_2w_2$ gives diagonal value 0 to $(i, 1) \neq 0$, also violating positive definiteness.

Shortcuts

- Check positivity on a convenient nonzero vector first; one failure disproves an inner product.
- To compute in the second position, conjugate scalar coefficients; this follows from the basic-properties proposition.
- Weighted coordinate products with strictly positive weights are positive definite because their diagonal values are sums of positive multiples of squared moduli.

Traps

- Over $\mathbb{C}$, the second position is generally not linear: $\langle 1, i\rangle = -i$ for the standard product on $\mathbb{C}$, whereas $i\langle 1, 1\rangle = i$.
- A nonnegative quadratic expression need not be positive definite: $(x_1 + x_2)^2$ vanishes at the nonzero vector $(1, -1)$.

§3 Norms and Cauchy–Schwarz

The question. How large can an inner product be relative to the two lengths? For $u = (1, 1)$ and $v = (1, -1)$ the inner product is 0; for $u = v = (1, 1)$ its modulus reaches the product of lengths, equal to 2.

Definition (Notes Definition 44; LADR 6.7)**.** A norm is a function $\|\cdot\| : V \to [0, \infty)$ with $\|v\| = 0$ exactly when $v = 0$, $\|av\| = |a|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$. For an inner product put $\|v\| = \sqrt{\langle v, v\rangle}$; the norm properties are proved below, using only this formula first.

Example (LADR 6.8)

The standard coordinate formula gives $\|z\| = (\sum_j |z_j|^2)^{1/2}$; the L^2 product gives $\|f\| = (\int_a^b |f|^2)^{1/2}$. For instance $(1, i)$ has norm $\sqrt{2}$, and $f(x) = x$ on $[0, 1]$ has norm $1/\sqrt{3}$.

Proposition (Notes Proposition 82; LADR 6.14: Cauchy–Schwarz)

$|\langle u, v\rangle| \leq \|u\|\|v\|$, with equality exactly when u, v are linearly dependent.

Where the proof comes from

Want. Bound a mixed inner product by two diagonal values.

Know. Every squared length is nonnegative.

First try. Positivity of the two diagonal values alone does not control the mixed term; use a third nonnegative diagonal value involving both vectors.

Key step. Expand the squared length of $u - av$ and choose a to cancel the mixed terms as far as possible.

Proof. 1. If $v = 0$, both sides vanish and the pair is dependent. Otherwise set $a = \langle u, v\rangle / \|v\|^2$ and $z = u - av$.
2. Expand using conjugate linearity: $0 \le \langle z, z\rangle = \|u\|^2 - \overline{a}\langle u, v\rangle - a\langle v, u\rangle + |a|^2\|v\|^2 = \|u\|^2 - |\langle u, v\rangle|^2/\|v\|^2$.
3. Multiply by $\|v\|^2 > 0$ and take nonnegative square roots to obtain the inequality.
4. Equality holds exactly when $\langle z, z\rangle = 0$, hence $z = 0$, so $u = av$. Conversely any such multiple, or a pair containing zero, has equality by direct substitution.

□

Proposition (Notes Proposition 81; LADR 6.9, 6.17)

The function induced by an inner product is a norm. In its triangle inequality, equality holds exactly when one vector is a nonnegative real multiple of the other.

Where the proof comes from

Want. Verify all norm axioms and identify equality in the triangle inequality.

Know. Positivity controls zero, and Cauchy–Schwarz bounds the mixed term in an expanded square.

Key step. Compare the square of $\|u + v\|$ with the square of $\|u\| + \|v\|$.

Proof. 1. Positive definiteness gives $\|v\| = 0$ exactly when $v = 0$. Sesquilinearity gives $\|av\|^2 = a\overline{a}\|v\|^2$, so $\|av\| = |a|\|v\|$.
2. Expand $\|u + v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2$. Since $\operatorname{Re} z \le |z|$ and Cauchy–Schwarz applies, this is at most $(\|u\| + \|v\|)^2$. Taking square roots proves the triangle inequality.
3. If both vectors are nonzero, equality requires equality in both bounds: dependence gives $u = cv$, and $\operatorname{Re}\langle u, v\rangle = |\langle u, v\rangle|$ gives c real and nonnegative. Such multiples do have equality; a zero vector also gives equality and is the zero multiple of the other.

□

Example (LADR 6.16: sum and integral forms)

For real a_j, b_j, $(\sum_j a_j b_j)^2 \le (\sum_j a_j^2)(\sum_j b_j^2)$. For continuous real f, g on $[-1, 1]$, $|\int_{-1}^{1} fg|^2 \le (\int_{-1}^{1} f^2)(\int_{-1}^{1} g^2)$. Both follow by applying Cauchy–Schwarz in the inner-product spaces already verified; equality means the two vectors or functions are dependent.

Bound an integral (Notes Example 67)

1. On $[0, \pi]$, take $f(x) = \sqrt{\sin x}$ and $g(x) = 1$, continuous real functions. Then $\|f\|^2 = \int_0^\pi \sin x \, dx = 2$ and $\|g\|^2 = \pi$.
2. Cauchy–Schwarz gives $\int_0^\pi \sqrt{\sin x} \, dx \leq \sqrt{2\pi} \approx 2.507$. Numerical integration gives about 2.396, consistent with the bound. The functions are not dependent, so the inequality is strict.

Shortcuts

- To bound a sum, identify it as an inner product; choose the second vector so its norm is simple.
- For equality, check dependence for Cauchy–Schwarz and a nonnegative real multiple for the triangle inequality.

Traps

- Complex dependence alone does not give triangle equality: $u = 1, v = i$ have $|u + v| = \sqrt{2} < 2 = |u| + |v|$.
- Equality in Cauchy–Schwarz includes zero vectors; requiring a nonzero scalar multiple would incorrectly exclude $u = 0, v \neq 0$.

§4 Parallelogram law, polarization and real angles

The question. Can every norm be obtained from an inner product? For the maximum norm $\|x\|_\infty = \max_j |x_j|$ on $\mathbb{R}^2$, both $e_1 + e_2$ and $e_1 - e_2$ have norm 1, so the proposed equality $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ would equate 2 and 4.

Proposition (Notes Proposition 83; LADR 6.21, 6A Exercise 28)

A norm is induced by an inner product if and only if it satisfies $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$. The converse is the cited Jordan–von Neumann theorem, as in the notes. When the inner product exists, it is uniquely recovered by

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4} \quad (F = \mathbb{R}),$$

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2}{4} \quad (F = \mathbb{C}).$$

Where the proof comes from

Want. Test whether a norm can come from an inner product and recover that product.

Know. Expanding squared lengths produces diagonal and mixed terms.

Key step. Add opposite-sign expansions to cancel mixed terms, or subtract them to isolate real and imaginary parts.

Proof of necessity and the formulas.

1. Expanding $\|u+v\|^2$ and $\|u-v\|^2$ and adding cancels the mixed terms, giving the parallelogram identity.
2. Subtracting gives $\|u+v\|^2 - \|u-v\|^2 = 4\,\mathrm{Re}\langle u, v\rangle$. This is the real formula.
3. Since $\langle u, iv\rangle = -i\langle u, v\rangle$, the analogous difference at iv is $4\,\mathrm{Im}\langle u, v\rangle$. Combining real and imaginary parts gives the complex formula and uniqueness.
4. Sufficiency for an arbitrary norm satisfying the identity is the explicitly cited Jordan–von Neumann theorem; it is not inferred from these expansions, which assume an existing inner product.

□

Example (Notes Chapter 17: maximum norm)

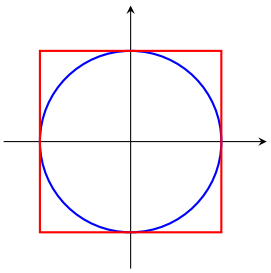
For F^n, $\|z\|_\infty = \max_j |z_j|$ is a norm. For $n \geq 2$, it is not induced by an inner product.

1. The maximum is nonnegative and zero exactly when every coordinate is zero; $\max_j |az_j| = |a| \max_j |z_j|$.
2. For each j, $|u_j + v_j| \leq |u_j| + |v_j| \leq \|u\|_\infty + \|v\|_\infty$. Taking the maximum proves the triangle inequality.
3. For $u = e_1, v = e_2$, the parallelogram left side is $1+1=2$ and the right side is $2+2=4$. Proposition 83 rules out an inducing inner product. In dimension 1 it is the usual absolute-value norm, so that case is different.

Definition (Notes Definition 45; LADR 6A Exercise 16)**.** For nonzero vectors in a real inner-product space, their angle is the unique $\theta \in [0, \pi]$ with $\cos\theta = \langle u, v\rangle / (\|u\|\|v\|)$. Cauchy–Schwarz puts the ratio in $[-1, 1]$; existence and uniqueness use the cited elementary inverse-cosine property on that interval.

Example (Notes Example 68)

For $u = (1, 0)$ and $v = (1, 1)$, $\langle u, v\rangle = 1$, $\|u\| = 1$ and $\|v\| = \sqrt{2}$, so $\cos\theta = 1/\sqrt{2}$ and $\theta = \pi/4$.



The standard inner-product norm $\|x\|_2 = (x_1^2 + x_2^2)^{1/2}$ has a unit circle; the maximum norm has a unit square. These give different lengths. The maximum norm fails the parallelogram law.

Shortcuts

- Test the parallelogram identity on coordinate vectors to disprove that a norm is induced by an inner product.
- Polarization recovers mixed inner products from lengths, with the complex signs fixed by first-position linearity.

Traps

- The angle definition requires nonzero vectors; at zero its denominator vanishes.
- The real cosine formula cannot directly define an angle for complex vectors: $\langle 1, i\rangle = -i$ is not real.

§5 Attention as inner-product scoring and weighted averaging

The question. How can vector inner products determine the coefficients of a linear combination? The vector $(1, 0)$ has dot products $2, 0, -1$ with $(2, 0), (0, 1), (-1, 0)$; these different real numbers can be converted into different positive coefficients.

Definition (Notes Chapter 17: query, key, value and softmax)**.** Let m, d_k, d_v be positive integers. Store pairs (k_j, ν_j) with $k_j \in \mathbb{R}^{d_k}$ and $\nu_j \in \mathbb{R}^{d_v}$. A query $q \in \mathbb{R}^{d_k}$ is compared with keys by scaled scores $s_j = \langle q, k_j\rangle/\sqrt{d_k}$. Softmax assigns weights $a_j = e^{s_j}/\sum_{\ell=1}^{m} e^{s_\ell}$. The attention output is $y = \sum_j a_j \nu_j$. Keys determine the weights; values are the vectors being combined, and may have a different dimension.

Proposition (Notes Chapter 17: stochastic weights)

Each softmax weight is positive and their sum is 1. Thus each output is a convex combination of values, meaning a linear combination with nonnegative coefficients summing to 1.

Where the proof comes from

Want. Normalize arbitrary real scores into weights.
Know. Real exponentials are strictly positive.
Key step. Divide every positive exponential by their common total.

Proof. 1. Every numerator and the denominator are positive, so $a_j > 0$.
2. Summing gives $\sum_j a_j = (\sum_j e^{s_j})/(\sum_\ell e^{s_\ell}) = 1$. Substitute the weights into the definition of y.

□

Compute the toy country–capital example (Notes Example 69)

1. Choose keys $k_1 = (2,0), k_2 = (0,1), k_3 = (-1,0)$ for France, Germany and Italy, and values $\nu_1 = e_{\text{Paris}}, \nu_2 = e_{\text{Berlin}}, \nu_3 = e_{\text{Rome}}$, the standard basis of $\mathbb{R}^3$. These are specified encodings for the example.
2. For $q = (1,0)$ and $d_k = 2$, the scores are $\sqrt{2}, 0, -1/\sqrt{2}$. Put $Z = e^{\sqrt{2}} + 1 + e^{-1/\sqrt{2}}$.
3. The output is $y = (e^{\sqrt{2}}/Z, 1/Z, e^{-1/\sqrt{2}}/Z) \approx (0.734, 0.178, 0.088)$ in that value basis. The first value has the greatest coefficient because the first score is largest.
4. In trained models the keys and values are learned internal vectors; the example's literal country and capital coordinates are not an assumption about their learned coordinates.

Proposition (Notes Chapter 17: angle and scale in a score)

For nonzero real query and key, $\langle q, k\rangle = \|q\|\|k\| \cos\theta$. For unit vectors this equals the cosine similarity $\cos\theta$; otherwise lengths also affect the score.

Where the proof comes from

Want. Identify what the inner-product score measures.
Know. The real angle was defined by its normalized inner product.
Key step. Rearrange that defining equation.

Proof. 1. Multiply the angle definition by the nonzero product $\|q\|\|k\|$ to obtain the formula.
2. If both lengths equal 1, the product is 1. For example $(1,0)$ with keys $(1,0)$ and $(2,0)$ has the same angle but scores 1 and 2 before scaling.

□

The scaling heuristic in the notes. Suppose the coordinates q_r, k_r are mutually independent random variables, each of mean 0 and variance 1. For this heuristic, write $\mathbb{E}X$ for the mean and $\operatorname{Var} X = \mathbb{E}[(X - \mathbb{E}X)^2]$. We cite elementary probability's factorization of expectations of independent variables and additivity of expectations.

1. Each product $q_r k_r$ has mean 0 and second moment $\mathbb{E}(q_r^2)\mathbb{E}(k_r^2) = 1$. Distinct products have mean-zero cross products by independence.
2. Expanding the squared sum gives $\operatorname{Var}(\sum_r q_r k_r) = d_k$, since the diagonal terms contribute 1 and the off-diagonal terms contribute 0.
3. Dividing by $\sqrt{d_k}$ makes that variance 1. This explains the chosen scale under the stated probabilistic assumptions; it does not assert independence for trained model coordinates.

Definition (Notes Chapter 17: learned maps and self-attention)**.** At position j let $x_j \in \mathbb{R}^{d_{\text{model}}}$ be its current representation. For one head, learned linear maps $W_Q, W_K : \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_k}$ and $W_V : \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_v}$ give $q_i = W_Q x_i$, $k_j = W_K x_j$, $\nu_j = W_V x_j$. Self-attention uses $s_{ij} = \langle q_i, k_j\rangle/\sqrt{d_k}$ and row-wise softmax weights a_{ij}, then $y_i = \sum_j a_{ij}\nu_j$. The query/key dimensions agree to make their inner product defined; the value dimension need not agree.

Definition (Notes Chapter 17: causal mask)**.** For generation from left to right, position i may use only positions $j \le i$. Normalize exponentials over those allowed positions: $a_{ij} = e^{s_{ij}} / \sum_{\ell \le i} e^{s_{i\ell}}$ for $j \le i$ and $a_{ij} = 0$ for $j > i$. Thus allowed weights are positive, excluded weights are zero, and each row still sums to 1. A matrix with nonnegative entries and row sums 1 is row-stochastic.

In the earlier Markov-chain calculation a fixed stochastic matrix moves probability vectors. Here the row-stochastic matrix is computed from the current input representations and combines value vectors. The weighting sums still use nonnegative coefficients summing to 1, with the row convention stated above.

Proposition (Notes Chapter 17: matrix calculation)

With columns $Q = (q_1 \ \cdots \ q_m)$, $K = (k_1 \ \cdots \ k_m)$ and $V_{\text{val}} = (\nu_1 \ \cdots \ \nu_m)$, the score matrix before masking is $S = Q^{\mathsf{T}}K/\sqrt{d_k}$. If $A = (a_{ij})$ is the row-stochastic attention matrix, the output matrix is $Y = V_{\text{val}}A^{\mathsf{T}}$.

Where the proof comes from

Want. Turn all position-wise sums into matrix products.

Know. Matrices store the vectors in columns, and multiplication sums over an intermediate index.

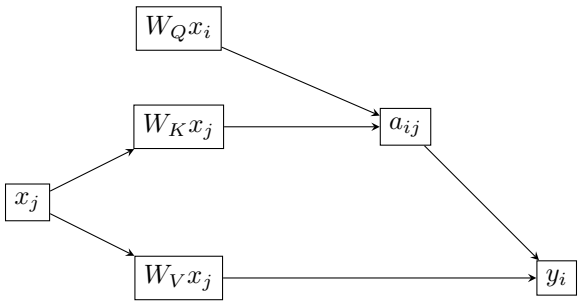
Key step. The (i, j) score is the dot product of column i of Q with column j of K; output column i uses row i of A.

Proof. 1. Q^T has size m-by-d_k, so $(Q^\mathsf{T}K)_{ij} = \sum_r (q_i)_r (k_j)_r = \langle q_i, k_j \rangle$. Division by $\sqrt{d_k}$ gives S_{ij}.
2. Column i of $V_{\text{val}}A^\mathsf{T}$ is $\sum_j \nu_j (A^\mathsf{T})_{ji} = \sum_j a_{ij}\nu_j = y_i$. Its size is d_v-by-m, as required.

□

Check a two-position causal calculation

1. Let $d_k = d_v = d_{\text{model}} = 1$, all three maps be the identity, and $x_1 = 0, x_2 = 1$. Then $S = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.
2. The causal mask gives $A = \left(\begin{smallmatrix} 1 & 0 \\ 1/(1+e) & e/(1+e) \end{smallmatrix}\right)$ and $V_{\text{val}} = (0\ 1)$. Thus $Y = V_{\text{val}}A^\mathsf{T} = (0\ e/(1+e))$.
3. More generally with $x_2 = t$, the second output is $f(t) = te^{t^2}/(1+e^{t^2})$. Since $f(2) = 2e^4/(1+e^4) \neq 2e/(1+e) = 2f(1)$, this attention map is not linear. The weighted-value step is linear for fixed A, but A itself depends on the input.



Query–key inner products determine the weights. The output combines the value vectors with those weights.

Context given in the notes.

1. A language model assigns probabilities to subsequent tokens, which may be words, word parts or punctuation. Initial vectors include token and position information; later representations also contain earlier computations.
2. The notes cite Bengio, Ducharme, Vincent and Jauvin's 2003 neural language model for learned word vectors, and Bahdanau, Cho and Bengio's 2014 translation attention for position-dependent weighting rather than one fixed-length sentence representation.
3. A transformer uses several heads and additional operations. The formulas here describe one head, and do not assert that individual learned coordinates or heads have a simple human-language interpretation.

Shortcuts

- Subtract the same constant from all scores in a row before exponentiating: the common factor cancels in numerator and denominator, so the weights stay unchanged. Choosing the maximum helps avoid large exponentials.
- Check matrix shapes: Q, K are d_k-by-m, A is m-by-m, and Y is d_v-by-m.
- Each attention row sums to 1, so transpose it when multiplying a matrix whose value vectors are columns.

Traps

- Masked weights are not all strictly positive: the two-position example has $a_{12} = 0$. Only allowed positions have positive weights.
- Attention is generally nonlinear despite the linear learned maps: the explicit function $f(t)$ above fails homogeneity.
- A query need not have the same dimension as a value: the toy example compares two-dimensional keys but returns three-dimensional values.

§6 Problem set

The question. Use trace identities, positivity tests and Cauchy–Schwarz before long calculations. For example $\operatorname{tr}\operatorname{diag}(1,-1)=0$ does not imply nilpotence.

HW9 P1. If $P^2=P$ on a finite-dimensional space over F, prove $\operatorname{tr} P=\operatorname{rank} P$, interpreted as $(\operatorname{rank} P)1_F$. You may use $V=\ker P\oplus\operatorname{range} P$.
Hint: Use an adapted basis and the trace definition; Week 6 proved the idempotent decomposition.

HW9 P2. On $\mathbb{C}^5$ let $m_T=(x-2)^2(x+1)$. Determine (a) all possible algebraic multiplicities of 2 and -1; (b) all traces; (c) a Jordan form realizing each trace.
Hint: Use Week 8's largest-block exponent and total-size formulas, then Proposition 79.

HW9 P3. (a) Prove every finite-dimensional complex nilpotent operator has trace zero. (b) Give a nonzero trace-zero operator which is not nilpotent. (c) Give two two-by-two matrices with nonzero $AB-BA$ and verify its trace is zero.
Hint: Use the nilpotent eigenvalue test and Propositions 79–80.

HW9 P4. Decide whether the following formulas on $\mathbb{R}^2$ are inner products: $x_1y_1+2x_2y_2$; $x_1y_1-x_2y_2$; $(x_1+x_2)(y_1+y_2)$. Justify every answer.
Hint: Apply positive weighted-product verification or test positive definiteness on e_2 and $(1,-1)$.

Orthogonality for HW9 P5(c); LADR 6.10. Two vectors are orthogonal if their inner product is zero. This definition suffices for the requested pair tests.

HW9 P5. On $\mathcal{P}_2(\mathbb{R})$ put $\langle p,q\rangle=\int_{-1}^{1}p(x)q(x)\,dx$. (a) Prove it is an inner product. (b) Compute the norms of $1,x,x^2$. (c) Determine which distinct pairs among these are orthogonal.
Hint: Use the continuous-function example and integrate the products; odd functions have zero integral on this symmetric interval.

HW9 P6. Prove $(a+b+c)^2\le 3(a^2+b^2+c^2)$ for real a,b,c, and determine exactly when equality holds.
Hint: Apply Cauchy–Schwarz to (a,b,c) and $(1,1,1)$.

HW9 P7 (optional). Prove the parallelogram identity in every real or complex inner-product space.
Hint: Expand both squared lengths as in Proposition 83.

Additional practice (LADR 8D Exercise 1). For fixed v,w in a finite-dimensional inner-product space, define $Tu=\langle u,v\rangle w$. Find $\operatorname{tr} T$.
Hint: In any basis write $w=\sum_j c_jb_j$ and sum the diagonal matrix entries; no special basis is needed.

Additional practice (LADR 6A Exercise 12). For positive real a,b,c,d, prove $(a+b+c+d)(1/a+1/b+1/c+1/d)\ge 16$ and determine equality.
Hint: Apply Cauchy–Schwarz to the square roots and their reciprocals.